

Ab Initio Electric-Field Gradients and Quadrupolar Resonances in Crystals

Practical scope, accuracy, and finite-temperature corrections

Abstract

Periodic density-functional theory (DFT) can calculate the electric-field-gradient (EFG) tensor at a nucleus and hence a quadrupolar coupling constant or nuclear-quadrupole-resonance (NQR) frequency. For small ordered crystals, the calculation is readily feasible on a workstation. Its limitation is normally predictive accuracy rather than computational cost: numerical convergence can be pushed to the percent level, whereas structural, finite-temperature, exchange-correlation, and near-nuclear reconstruction errors often leave absolute couplings uncertain by roughly 10–30%, with much larger relative errors near symmetry-forced zeros or in dynamically disordered systems. Finite-temperature values are obtained most systematically by averaging EFG *tensors* over a vibrational or molecular-dynamics ensemble; there is no generally reliable scalar correction that maps a single 0 K result to room temperature.

Table 1: Open-source tools for crystalline EFG and quadrupolar calculations.

Tool	Electronic-structure treatment	EFG route and output	Practical role
ABINIT	Plane waves, PAW	nucefg 2; reports the tensor, principal values/axes, V_{zz} , η , and, with quadmom , C_Q in MHz [1]	Direct and comparatively simple EFG-to- C_Q workflow; use PAW, not norm-conserving pseudopotentials
Quantum ESPRESSO + QE-GIPAW	Plane waves, USPP/PAW	gipaw.x EFG calculation for periodic systems [2]	Natural choice in an existing QE workflow; requires suitable GIPAW-compatible datasets
Elk	All-electron, full-potential LAPW	Task 115; tensor and eigenvalues written to EFG.OUT [3]	Valuable all-electron benchmark, especially for heavy atoms or PAW validation
CP2K	Gaussian/plane-wave; GPW or GAPW	ELECTRIC_FIELD_GRADIENT print section [4]	Useful with CP2K dynamics; GAPW is preferable for core-sensitive observables
Soprano	Post-processing only	Diagonalizes EFG tensors and evaluates V_{zz} , C_Q , and NQR quantities [5]	Analysis and convention handling; it does not calculate the electronic structure

1 Definitions and spectroscopic observables

At nuclear site n , the EFG is the Hessian of the electrostatic potential Φ ,

$$V_{\alpha\beta}^{(n)} = \left. \frac{\partial^2 \Phi(\mathbf{r})}{\partial x_\alpha \partial x_\beta} \right|_{\mathbf{r}=\mathbf{R}_n}. \quad (1)$$

Away from a charge singularity the tensor is symmetric and traceless. In its principal-axis system, use the common ordering

$$|V_{zz}| \geq |V_{yy}| \geq |V_{xx}|, \quad \eta = \frac{V_{xx} - V_{yy}}{V_{zz}}, \quad 0 \leq \eta \leq 1. \quad (2)$$

For isotope-specific nuclear quadrupole moment Q , the coupling constant is

$$C_Q = \frac{eQV_{zz}}{h}. \quad (3)$$

The sign and principal-axis ordering must be checked when comparing programs or literature conventions. In frequency units, one standard principal-axis quadrupolar Hamiltonian is

$$\frac{\hat{H}_Q}{h} = \frac{C_Q}{4I(2I-1)} \left[3\hat{I}_z^2 - I(I+1) + \frac{\eta}{2} (\hat{I}_+^2 + \hat{I}_-^2) \right]. \quad (4)$$

NQR frequencies are differences between eigenvalues of Eq. (4). For $I = 3/2$,

$$\nu_{\text{NQR}} = \frac{|C_Q|}{2} \sqrt{1 + \frac{\eta^2}{3}}. \quad (5)$$

Thus, except where the η dependence is unusually important, the fractional error in an NQR frequency approximately follows that in V_{zz} and C_Q . In high-field NMR, first-order quadrupolar splittings scale as C_Q , while second-order shifts and broadenings scale approximately as C_Q^2/ν_L ; a 10% error in C_Q therefore produces about a 20% error in a purely second-order contribution.

2 Feasibility for simple crystals

Static EFG calculations for NaF, NaNO₃, KNO₃, and similarly small unit cells are workstation-scale. A modern machine with roughly 8–16 CPU cores and 32–64 GB of memory is normally adequate for structural relaxation, a converged PAW ground state, and the EFG. A single calculation may take minutes to hours; the full cutoff and k -mesh convergence study is often the larger cost. ABINIT notes that its EFG evaluation adds little to the cost of the converged ground-state calculation [1].

Ideal rocksalt NaF is an important caveat: cubic site symmetry forces the EFG to zero. It is therefore a useful numerical symmetry check, but not a useful nonzero-frequency benchmark. Nitrate crystals remain small enough for routine static DFT, although nitrate orientation, polymorphism, internal coordinates, and librational motion can dominate the result. High-performance computing becomes attractive for large defect/disorder supercells, hybrid functionals, long ab initio molecular dynamics (AIMD), dense snapshot sampling, or path-integral molecular dynamics (PIMD), rather than for the basic primitive-cell EFG itself.

3 Accuracy and scientific utility

A useful separation is between *numerical convergence* and *physical-model error*. Cutoff, augmentation-grid, k -point, and self-consistency errors can often be reduced until V_{zz} changes by only 1–2%. This does not imply comparable agreement with experiment. The EFG is a second derivative of the potential at the nucleus and is unusually sensitive to small structural distortions, the exchange–correlation approximation, PAW reconstruction or basis completeness, semicore treatment, magnetism, relativity, the adopted quadrupole moment Q , and finite-temperature motion [6, 7].

A recent ²⁷Al benchmark reported raw mean errors of about 18% for CASTEP and 24% for Quantum ESPRESSO under a common semilocal-DFT protocol; calibrated correlations generally reduced average errors to roughly 10–20%. The asymmetry parameter showed mean absolute errors of about 0.13–0.16, whereas orientation-related quantities were more robust, with an average error near 5% [8]. A broader comparison covering 38 experimental EFGs reported a mean absolute percentage difference of 28.9% [9]. Percentage errors become ill-conditioned near cubic or nearly cubic sites: a small absolute residual divided by a nearly zero experimental EFG can look enormous, and small lattice distortions can even reverse the sign or principal-axis ordering.

For planning purposes, not as universal confidence intervals, the following ranges are realistic:

Situation	Typical absolute- C_Q or frequency error
Numerical convergence of a fixed model	approximately 1–2%
Favorable, rigid, ordered main-group crystal	approximately 5–10%
Routine uncalibrated semilocal DFT/PAW prediction	approximately 10–30%
Near-symmetric, mobile, disordered, wrong-phase, or poorly described system	50% or more is possible

Consequently, a raw DFT number is often a weak way to narrow a completely blind frequency scan: a $\pm 20\%$ interval may reduce the search range by only a modest factor. The method is more persuasive when used comparatively or inversely: predicting the number of inequivalent sites, tensor symmetry and orientation, assigning already

observed lines, discriminating polymorphs or defect models, and testing whether a structural model is consistent with experiment. System-specific calibration can also be effective; for example, a molecular correction scheme reduced the root-mean-square error of a benchmark set of ^{17}O couplings by 31% [10].

4 Finite-temperature treatment

There is no generally reliable scalar law $C_Q(T) = f(T)C_Q(0)$. The well-founded object is the thermodynamic average of the EFG tensor. Formally, for the nuclear Hamiltonian $\hat{H}_{\text{nuc}}$,

$$\langle \mathbf{V} \rangle_T = \frac{\text{Tr} \left[e^{-\beta \hat{H}_{\text{nuc}}} \mathbf{V}(\hat{\mathbf{R}}) \right]}{\text{Tr} \left[e^{-\beta \hat{H}_{\text{nuc}}} \right]}, \quad \beta = (k_B T)^{-1}. \quad (6)$$

In the classical configurational limit,

$$\langle \mathbf{V} \rangle_T = \frac{\int d\mathbf{R} \mathbf{V}(\mathbf{R}) e^{-\beta U(\mathbf{R})}}{\int d\mathbf{R} e^{-\beta U(\mathbf{R})}}. \quad (7)$$

Practical approximations form a hierarchy.

1. **Experimental structure at the measurement temperature.** Recompute the EFG using measured lattice constants and internal coordinates. This cheaply includes thermal expansion and any resolved static distortion, but not instantaneous vibration or libration.
2. **Harmonic or quasi-harmonic phonon sampling.** Draw normal-mode amplitudes from the quantum thermal distribution. For a mass-weighted normal coordinate q_λ ,

$$\langle q_\lambda^2 \rangle_T = \frac{\hbar}{2\omega_\lambda} \coth \left(\frac{\hbar\omega_\lambda}{2k_B T} \right), \quad (8)$$

which includes zero-point motion. Quasi-harmonic calculations add volume expansion. This is a good cost–accuracy compromise for rigid crystals away from phase transitions, but harmonic phonons can misestimate EFG trends when anharmonicity is important, as demonstrated for Zn and Cd [11, 12].

3. **AIMD at the target temperature.** Generate an NVT or NPT trajectory, select statistically independent snapshots, and evaluate an EFG tensor for each. AIMD captures anharmonic vibration, molecular-ion libration, and local coordination changes. In NaNO_2 , finite-temperature DFT-MD improved the mean ^{14}N and ^{17}O EFGs and revealed a broad, coordination-dependent ^{23}Na distribution absent from a static calculation [13].
4. **PIMD.** Represent each nucleus by a ring polymer to include nuclear quantum effects. This is most relevant at low temperature, for light atoms, or where zero-point motion strongly affects the local geometry; PIMD-based EFG averaging has been demonstrated for molecular solids [14].

If snapshots are reported in different coordinate frames, first transform every EFG into the same fixed crystal or laboratory frame—without aligning away the physical molecular reorientation—and then average,

$$\bar{\mathbf{V}}(T) = \sum_s w_s \mathbf{R}_s \mathbf{V}^{(s)} \mathbf{R}_s^\top, \quad \sum_s w_s = 1, \quad (9)$$

and only then diagonalize $\bar{\mathbf{V}}$ to obtain $V_{zz}(T)$, $\eta(T)$, $C_Q(T)$, and the transition frequencies. Directly averaging sorted $|V_{zz}|$ values, $|C_Q|$, or nonlinear frequencies can be wrong because principal axes rotate, eigenvalues can exchange labels, and diagonalization is nonlinear.

Equation (9) describes the fast-motion limit. If jumps, reorientations, ion hopping, or domain changes are slow or comparable to the quadrupolar evolution timescale, the experiment need not show one averaged resonance: separate lines, exchange broadening, distributions, or motional narrowing may occur. Spectral dynamics must then be modeled explicitly rather than compressed into one $C_Q(T)$.

A useful composite strategy is to calculate an accurate static baseline and a cheaper finite-temperature correction with the same lower-level method at both endpoints,

$$\mathbf{V}_{\text{best}}(T) \approx \mathbf{V}_{\text{high}}(0) + [\bar{\mathbf{V}}_{\text{low}}(T) - \mathbf{V}_{\text{low}}(0)]. \quad (10)$$

The temperature shift may benefit from error cancellation, although this is not guaranteed and does not remove uncertainty in the underlying functional, pseudopotential, isotope quadrupole moment, phase, or sampling.

5 Recommended decision workflow

1. Identify the correct polymorph, isotope, measurement temperature, and whether the site is symmetry-forced to have zero EFG.
2. Prefer the experimental structure at that temperature when the goal is comparison with experiment; separately test a relaxed 0 K geometry to diagnose structural sensitivity.
3. Use a PAW or all-electron method, and converge the wave-function cutoff, augmentation grid, k mesh, semicore treatment, and tensor components—not merely the total energy.
4. Establish the static uncertainty by comparing plausible functionals/datasets or by benchmarking against a chemically close reference. Do not interpret a 1% numerical convergence threshold as 1% predictive accuracy.
5. Add finite-temperature physics in increasing order of cost: measured geometry, phonon/quasi-harmonic sampling, AIMD, then PIMD where justified. Average tensors in a consistent frame and report statistical uncertainty and autocorrelation-aware effective sample size.
6. Use the result primarily as a structural discriminator or line-assignment aid unless system-specific validation shows that absolute frequencies are reliable to the precision required by the experiment.

Bottom line. Small-crystal EFG calculations are computationally easy, but high-accuracy blind frequency prediction is not. Static semilocal DFT commonly leaves 10–30% uncertainty in absolute couplings. Finite-temperature ensemble averaging is the established correction framework and can remove a dominant bias in dynamically active crystals, but it is not a universal route from a 0 K result to few-percent NQR accuracy.

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